MIL-R-23417/25A(AS)

5 November 1968

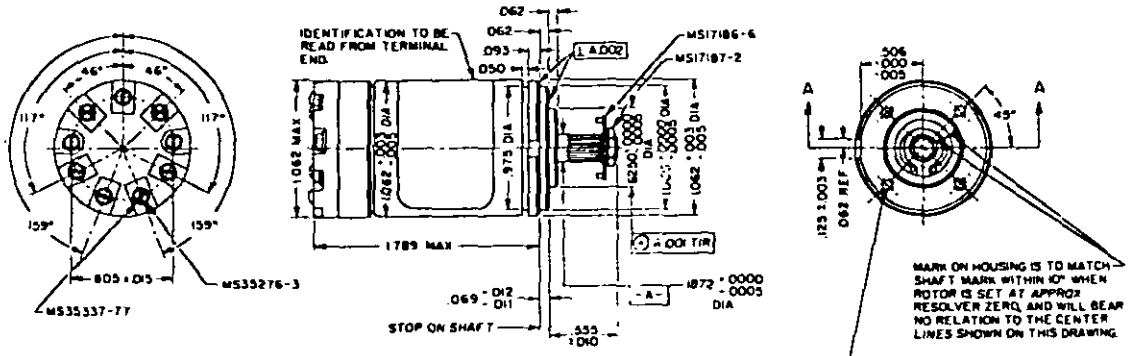
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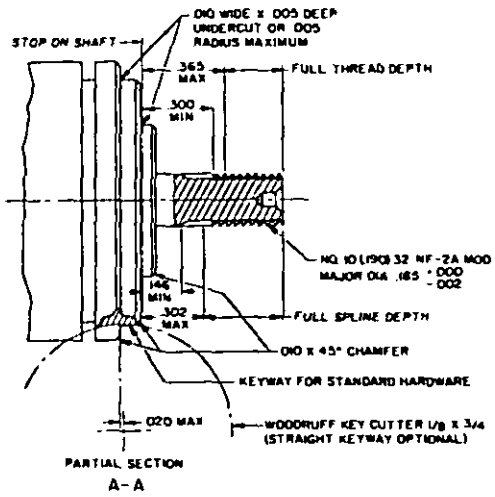
MIL-R-23417/25(WP)

13 January 1966

MILITARY SPECIFICATION SHEET
RESOLVER TRANSMITTER, TYPE 9V-11R41N40a

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of MIL-R-23417.





SPLINE DATA	
TEETH	21
PITCH	120
PITCH DIAMETER	.1750 +.0000 -.0020
OUTSIDE DIAMETER	.1872 +.0000 -.0005
PRESSURE ANGLE	20
ROOT DIAMETER MAXIMUM	.155
TOOTH FORM	FULL DEPTH INVOLUTE

NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. *SPLINE SHALL BE INSPECTED BY GO, NO-GO SPLINE RING* GAGE. MS90406-1.

NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE *SHAFT, THEN THE HOUSING* ROTATED.
2. RUNOUT OF FEATURE [-A-] (SMOOTH PORTION) OF SHAFT SHALL NOT EXCEED .0008 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED *DURING THE REVERSAL OF THE FOLLOWING LOADS*
   END PLAY - 1 POUND
   RADIAL PLAY - 1/2 POUND
4. THE END CAP AND TERMINAL BLOCK SHALL BE SECURED BY SCREWS OTHER THAN THE TERMINAL SCREWS. RECESSES SHALL ALLOW TERMINAL LUGS MS17182-2 TO BE INSTALLED WITH THE BARREL IN-BOARD OR OUTBOARD AND PREVENT THE LUGS FROM TURNING.
5. THE FOLLOWING LIST OF HARDWARE MS35276-3 MACHINE SCREW (7 REQUIRED), MS35337-77 LOCK WASHER (7 REQUIRED), MS17182-2 TERMINAL LUG (7 REQUIRED), MS17186-6 DRIVE WASHER (1 REQUIRED), MS17187-2 NUT (1 REQUIRED), SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS, THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT RESOLVER PACKAGE.
6. COLLECTOR RING MATERIAL SHALL BE 69 GOLD, 06 PLATINUM AND 25 SILVER.
7. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL) A SAMPLE OF THE RESOLVERS TO BE FURNISHED *UNDER THIS SPECIFICATION* SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN MIL-R-23417.

FSC 5998

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TABLE I

RESOLVER TRANSMITTER, TYPE 9V-11R41N40a
PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

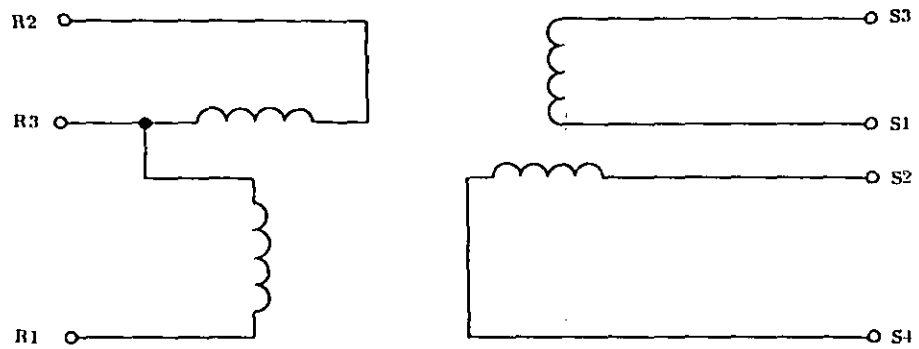
REQUIREMENTS	VALUE	UNIT	TOLERANCE
Rated frequency	4000	cps	±1%
Input voltage	9.0	Volts	±10%
Test voltage	9.0	Volts	±1%
Input current	2.6	Milliamps	Nominal
Input power	0.01	Watts	Nominal
Angular error	7.0	Minutes	Maximum
Transformation ratio	0.497	----	±4%
Null voltage:			
Total	20	Millivolts	Maximum
Fundamental	15	Millivolts	Maximum
Temperature rise	20	Degrees (C)	Maximum
Phase shift (Input to output secondary open circuit)	0	Degrees	±1.5
Friction torque:			
Room temperature	10	Gm-Cm	Maximum
-54 to 85° C	20	Gm-Cm	Maximum
Impedance:			
Zro	3720-4467	Ohms	Min.-Max.
Zss	121-145	Ohms	Min.-Max.
Impedance angle:			
Zro	68.0-73.0	Degrees	Min.-Max.
Zss	65.0-69.0	Degrees	Min.-Max.

Notes:

1. The Resolver Transmitter identified herein shall conform to all requirements and shall be subjected to all tests delineated in MIL-R-23417.

2. The type 9V-11RX41N40a Resolver was formerly known as the 9V-11RX40a, also as the 11R10N40 Mk 8 Mod O.

3. Sampling, inspection and preparation for delivery shall be in accordance with MIL-R-23417.



9V/4.5V
CONNECTION DIAGRAM

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Custodian:

Navy-AS

Preparing activity:

Navy-AS

Project 5990-N004

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